HazeChain

Satellite-verified peat fire evidence anchored on Ethereum, and a smart contract that pays communities only when the satellites confirm their land stayed fire-free.

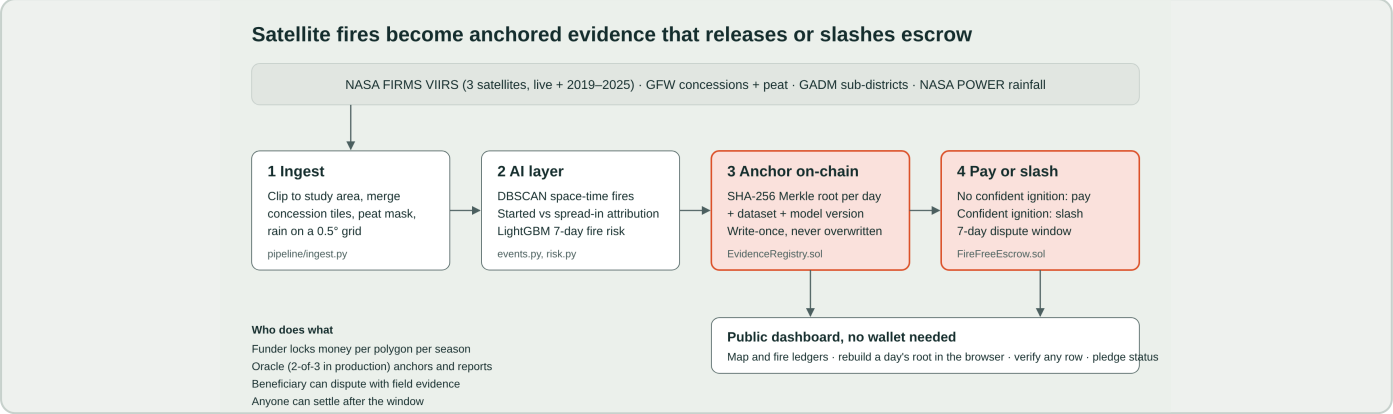
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Track: Climate Data & Environmental Monitoring
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shermaineyap.github.io/hazechain-IEEEHack

The problem, this week

From 17 to 24 October 2026, NASA satellites recorded 58,466 fire detections in one corner of Central Kalimantan. The haze has hit Indonesia, Malaysia and Singapore since September. Satellites see every fire; what fails is **attribution** (fires cross boundaries and evidence is contested) and **incentives** (paying villages to stay fire-free works, but runs on one company's paperwork). Each hectare of drained peat that burns emits about **800 tCO_2e** (IPCC 2013 Wetlands Supplement).

What HazeChain does

- **AI** clusters pixels into fires, finds each ignition and decides whether a fire *started* in a polygon or *spread in*; LightGBM forecasts next week's risk.
- **EvidenceRegistry** stores one SHA-256 Merkle root per day, write-once, with the model version.
- **FireFreeEscrow** pays a village or company when no confident ignition is reported, and slashes to a suppression pool when one is, after a dispute window.
- **Anyone** can rebuild a root in the browser and verify a row on-chain, with no wallet.



79%
of this week's fire pixels on drained peat; only 25% inside concessions

152 / 260
concessions fire-free all week, the land managers a fund should pay

0.83 vs 0.68
risk-model ROC-AUC vs naive baseline, 2025 season held out (0.87 vs 0.83 in El Niño 2023)

14 / 14
contract tests: tamper, pay, slash, low confidence, dispute, refund, double-settle

Live pledges on Sepolia

Polygon	Satellites show	Outcome
Karau Kuala sub-district	No fire, 84% peat	Paid to village
Concession GFW-OP-817	No fire, 99% peat	Paid
Concession GFW-OP-877	5 confident ignitions	Slashed to pool
Concession GFW-OP-799	1 ignition at 0.71, 2 spread in	Disputed

Demo pledges with test ETH; not commitments by, or accusations against, the named organisations.

Why blockchain, impact, adoption

A village, a funder and a regulator who distrust each other need one record nobody can edit, and payment that happens automatically on the outcome. APRIL's Fire Free Village pilots cut burnt area 90% for about US$30,000 per village; keeping 1,000 ha of peat from burning at that cost is under **US$0.04 per tCO_2e**. Users: village fire-care groups, plantations, commodity buyers (EUDR, RSPO), Indonesia's environment ministry, haze regulators, results-based climate finance. Next: a 2027 regency pilot with a 2-of-3 oracle set. NASA FIRMS is global, so the same pipeline covers every peatland.

Security and data integrity

Write-once roots, oracle allow-list, reentrancy guard, no double settlement, funder refund if the oracle disappears, dispute window, spread-in fires never count, 0.70 confidence floor. Residual risk is the single oracle; production uses a staked multi-party oracle and multisig arbiter.

Honest limits

Concession maps are from 2012. Attribution confidence is a published heuristic, not a trained classifier, because no labelled ground truth exists yet. A 7-day window censors long-burning fires, which are flagged and down-weighted. VIIRS misses small, cloud-covered and smouldering fires.

Data: NASA FIRMS VIIRS 375 m (S-NPP, NOAA-20, NOAA-21); Global Forest Watch concessions and peatlands; NASA POWER; GADM 4.1; IPCC 2013 Wetlands Supplement Tables 2.6–2.7. Code: github.com/ShermaineYap/hazechain-IEEEHack. Contracts: REGISTRY_ADDRESS, ESCROW_ADDRESS (Ethereum Sepolia).